Two-color excitation with focused optical pulses: background, replication, and how it connects to the current work

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The new direction: two beams of different color, each tightly focused, excite a sample only where they overlap in space and time. This document builds the physics from Maxwell's equations up to that signal (Part I), reports the replication of the three low-NA papers the work starts from (Part II), and connects both to the Richards–Wolf pulsed-focus work already done (Part III). Every citation was checked against its DOI record (Crossref) or ISBN.

I. Background from first principles

1 From Maxwell's equations to the Helmholtz equation

In a source-free, linear, isotropic medium of refractive index n, Maxwell's equations give $\nabla \cdot \mathrm{E} = 0$ and $\nabla^2\mathrm{E} - (n^2/c^2)\,\partial_t^2\mathrm{E} = 0$ [1, 3]. For monochromatic light, $\mathrm{E}(\mathrm{r}, t) = \mathrm{Re}[\mathrm{E}(\mathrm{r})e^{-i\omega t}]$, and the time dependence drops out:

$$\nabla^2\mathrm{E} + k^2\mathrm{E} = 0, \qquad \nabla \cdot \mathrm{E} = 0, \qquad k = \frac{n\omega}{c} = \frac{2\pi n}{\lambda}.$$

The first is the Helmholtz equation; the second, transversality, is what makes the field a vector problem. Detectors and molecules respond to the cycle-averaged intensity, $I \propto |\mathrm{E}|^2$.

2 Plane waves and the angular spectrum

The simplest solution is a plane wave, $\mathrm{E} = \mathrm{E}_0\, e^{i\mathrm{k}\cdot\mathrm{r}}$ with $|\mathrm{k}| = k$. Transversality forces $\mathrm{k} \cdot \mathrm{E}_0 = 0$: *the field of every plane wave is perpendicular to its own direction*. Any source-free field in a half-space can be written as a superposition of such plane waves, its angular spectrum [2, 3]. A focus is exactly this: many plane waves, arriving from many directions, adding up at one point. Because each wave must stay transverse to its own direction, waves arriving at large angles necessarily carry a component along the optical axis.

3 A lens and its focus: the Debye integral

A lens turns an incoming plane wave into a spherical wave converging on the focus. Huygens' principle, made quantitative by Kirchhoff, propagates that wave to the focal region [1, 2]. Debye [5] showed that when the aperture is many wavelengths across and the Fresnel number $a^2/\lambda f$ is large (the condition Lim & Saloma state, [14] p. 122), the focal field is a superposition of plane waves whose directions fill the cone the aperture subtends at the focus, each weighted by the field on the aperture. Wolf put this in integral form

[6]. For a scalar field and a pupil that depends only on the angle ϑ from the axis,

$$U(r,z) = \int_0^{\alpha}\int_0^{2\pi} a(\vartheta)\, e^{ik(r\sin\vartheta\cos(\varphi-\psi)+z\cos\vartheta)} \sin\vartheta\, d\varphi\, d\vartheta$$
$$= 2\pi \int_0^{\alpha} a(\vartheta)\, J_0(kr\sin\vartheta)\, e^{ikz\cos\vartheta} \sin\vartheta\, d\vartheta,$$

where $\sin\alpha = \mathrm{NA}/n$, $\sin\vartheta\, d\vartheta\, d\varphi$ is the solid-angle element, and the azimuthal integral is the Bessel identity

$$\int_0^{2\pi} e^{iv\cos(\varphi-\psi)}\, d\varphi = 2\pi J_0(v).$$

4 The aplanatic lens: where $\sqrt{\cos\vartheta}$ comes from

A well-corrected (aplanatic) lens obeys the sine condition: a ray entering at height h leaves toward the focus at the angle ϑ with $h = f\sin\vartheta$ (Fig. 1) [1, 7]. Energy conservation then fixes the amplitude on the reference sphere. The power through a pupil ring, $|E_{\mathrm{in}}|^2\, 2\pi h\, dh$, must equal the power through the matching band of the sphere, $|E_{\mathrm{sph}}|^2\, 2\pi f^2 \sin\vartheta\, d\vartheta$. With $dh = f\cos\vartheta\, d\vartheta$,

$$|E_{\mathrm{sph}}|^2 = |E_{\mathrm{in}}|^2\cos\vartheta \quad\Longrightarrow\quad a(\vartheta) = A(\vartheta)\sqrt{\cos\vartheta},$$

where $A(\vartheta)$ is the input beam profile: 1 for uniform illumination, $\exp(-\alpha_t \sin^2\vartheta/\sin^2\alpha)$ for the truncated Gaussian of the current work.

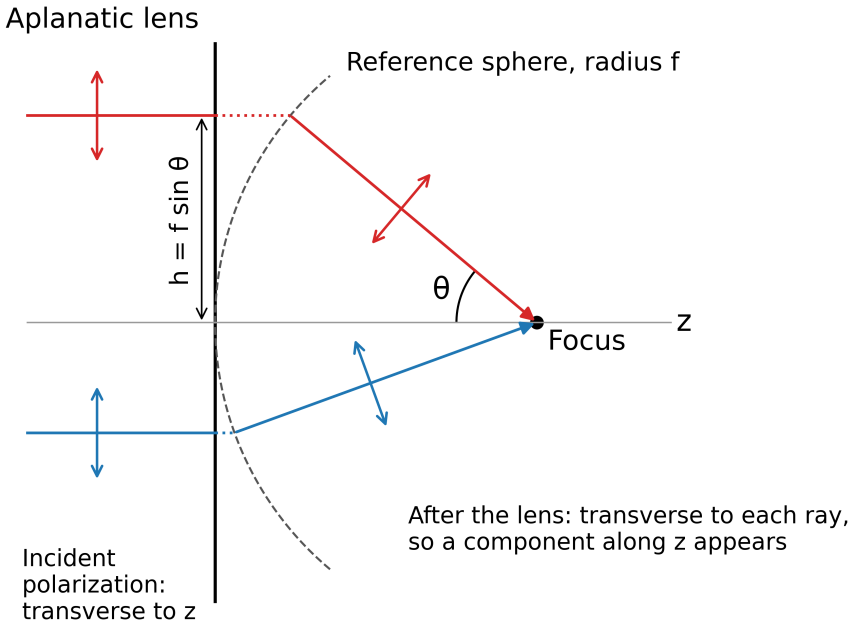


Figure 1: Aplanatic focusing. Each ray leaves the lens at height $h = f\sin\vartheta$ and travels along a radius of the reference sphere. Its polarization, transverse to z before the lens, must be transverse to the ray afterwards, so rays at large ϑ carry a field component along z.

5 Polarization through the lens: Richards–Wolf

Richards and Wolf [7] carried the vector field through the aplanatic lens. For x-polarized input, the ray at (ϑ, φ) reaches the reference sphere with unit field [3]

$$\hat{\mathrm{e}}(\vartheta, \varphi) = \big(\cos\vartheta\cos^2\varphi + \sin^2\varphi,\ (\cos\vartheta - 1)\sin\varphi\cos\varphi,\ -\sin\vartheta\cos\varphi\big).$$

The component along the ray's radial (meridional) direction is tilted with the ray; the azimuthal component is unchanged. Rewriting the x component as $\frac{1}{2}(1 + \cos\vartheta) - \frac{1}{2}(1 - \cos\vartheta)\cos 2\varphi$ and integrating each term over φ with $\int_0^{2\pi} \cos(m\varphi)\, e^{iv\cos(\varphi-\psi)} d\varphi = 2\pi i^m J_m(v)\cos(m\psi)$ gives the focal field [7, 16]:

$$E_x = -i(I_0 + I_2\cos 2\psi), \qquad E_y = -iI_2 \sin 2\psi, \qquad E_z = -2iI_1\cos\psi,$$

$$I_m = \int_0^{\alpha} A(\vartheta)\sqrt{\cos\vartheta}\,\sin\vartheta\, g_m(\vartheta) J_m(kr\sin\vartheta)\, e^{ikz\cos\vartheta}\, d\vartheta, \qquad g_0 = 1 + \cos\vartheta,\ g_1 = \sin\vartheta,\ g_2 = 1 - \cos\vartheta.$$

I_0 is the familiar focusing term; I_2 breaks the circular symmetry of the spot; I_1 is the longitudinal field E_z. This is the field the current work computes (Romallosa replication, 472.8 nm checkpoint).

6 The low-NA limit: the model the two-color papers use

For a small cone, $\vartheta \ll 1$: $\sin\vartheta \approx \vartheta$, $\cos\vartheta \approx 1 - \vartheta^2/2$, so $g_0 \to 2$, $g_1, g_2 \to 0$ (E_y and E_z vanish), $\sqrt{\cos\vartheta} \to 1$, and the phase becomes quadratic. With $\rho = \vartheta/\alpha$,

$$U(u, v) \propto e^{ikz}\, 2\int_0^1 J_0(v\rho)\, e^{-iu\rho^2/2}\rho\, d\rho, \qquad u = k\Big(\frac{\mathrm{NA}}{n}\Big)^2 z, \quad v = \frac{2\pi}{\lambda}\,\mathrm{NA}\, r,$$

the scalar field expressed by Lommel functions [1]. Its focal-plane profile is the Airy pattern and its axial profile $\mathrm{sinc}^2(u/4)$. All three two-color papers use it [13, 14, 15]. Figure 2 shows what it misses.

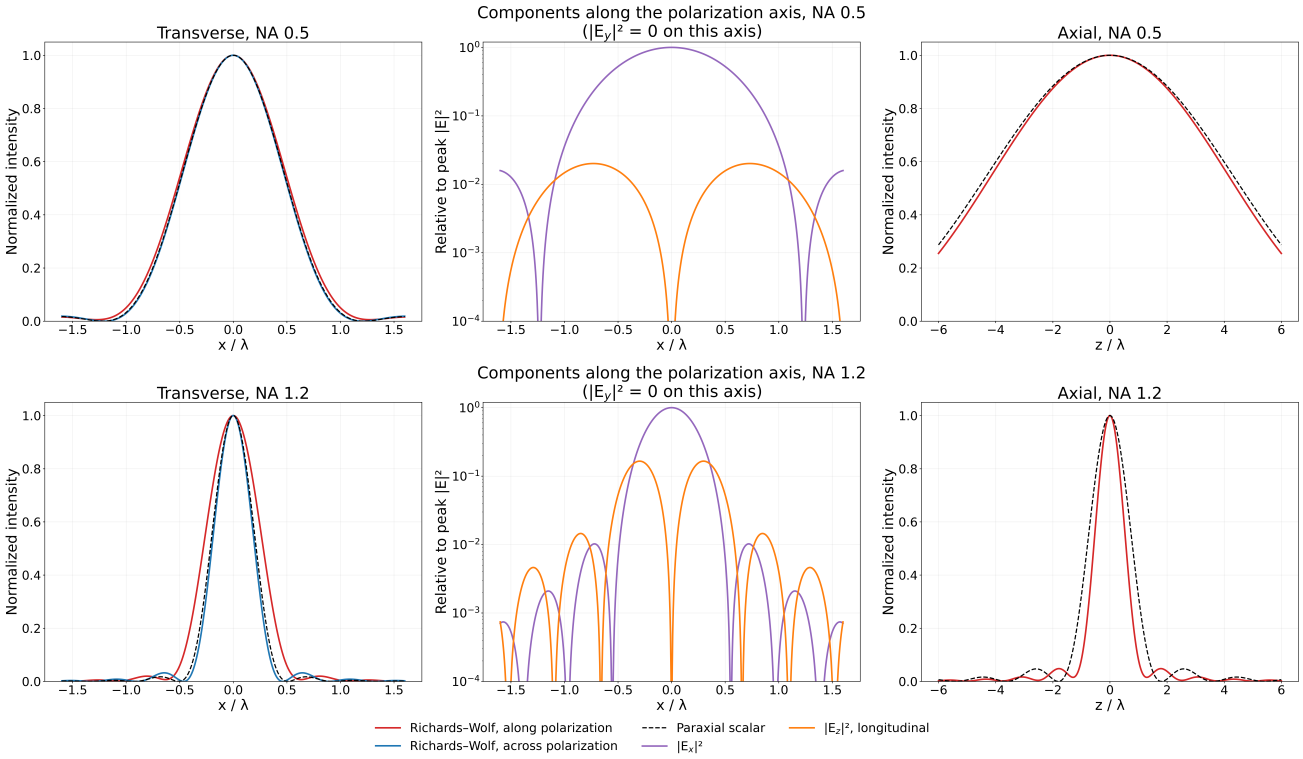


Figure 2: Richards–Wolf against the paraxial scalar field, 750 nm, $n = 1.3$, uniform input, x-polarized. At NA 0.5 the two nearly coincide transversely; E_z is 2% of the peak. At NA 1.2 the spot is elongated along the polarization, E_z reaches 16% of the peak off axis, and the scalar model is visibly wrong. Axially the Richards–Wolf focus is 4% shorter than the paraxial one at NA 0.5 and much shorter at NA 1.2, because the exact phase $kz\cos\vartheta$ departs from its quadratic approximation.

The current work puts the 1% focal-width departure between scalar and vector theory at NA ≈ 0.45, so the papers' NA 0.5 sits at the edge of their model's validity.

7 Light and matter: one, two, and two-color excitation

One photon. A molecule absorbs a photon of energy $\hbar\omega$ at a rate proportional to $|E|^2 = I$. Integrated over any plane through the beam, $\int I\,dA$ is the beam power, the same in every plane: one-photon fluorescence comes from the whole cone, not only the focus (Fig. 3).

Two photons. Göppert-Mayer showed that two photons can be absorbed together through a short-lived virtual state, at a rate proportional to I^2 [8]. Now $\int I^2\,dA$ falls away from the focus, so excitation is confined to the focal volume. This is the basis of two-photon microscopy [9].

Two colors. The two photons can come from different beams, $\hbar\omega_1 + \hbar\omega_2 = \hbar\omega_e$, i.e. $1/\lambda_e = 1/\lambda_1 + 1/\lambda_2$ (Fig. 3); two-color two-photon excitation has been observed experimentally [10]. The rate is proportional to the product of the two intensities [13]:

$$F_{2c}(P) = \kappa\, I_1(P)\, I_2(P).$$

It is zero wherever either beam is absent, so fluorescence exists only where the two foci overlap. Each beam alone can still drive ordinary two-photon absorption if its own photons suffice, which is why probes are

chosen to suppress it [15].

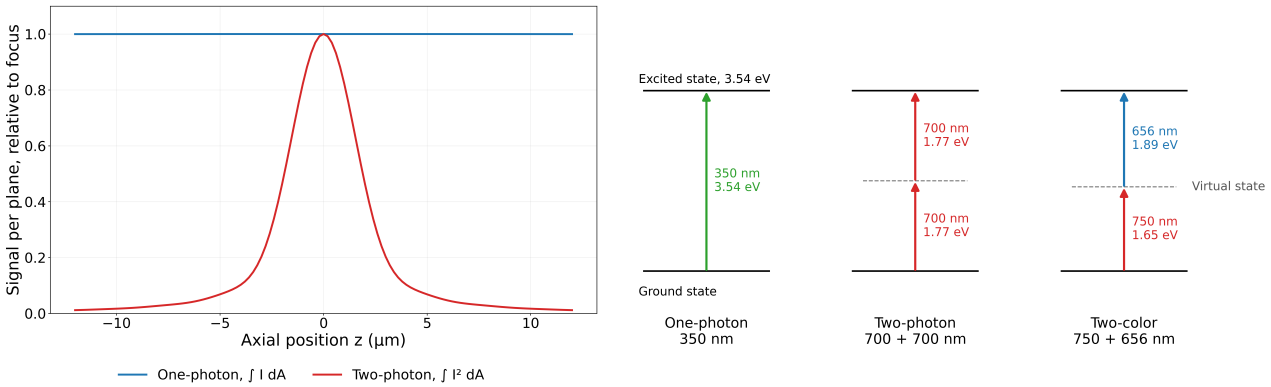


Figure 3: Left: signal per plane against distance from focus (paraxial, NA 0.5, 700 nm). One-photon excitation is the same in every plane; two-photon excitation is confined to the focus. Right: the three transitions to the same 3.54 eV state.

8 Why two colors do not interfere and two photons of one color do

With both beams present, the total field is $\mathrm{E}_1 e^{-i\omega_1 t} + \mathrm{E}_2 e^{-i\omega_2 t}$, and

$$|\mathrm{E}|^2 = |\mathrm{E}_1|^2 + |\mathrm{E}_2|^2 + 2\,\mathrm{Re}\big[\mathrm{E}_1 \cdot \mathrm{E}_2^*\, e^{-i(\omega_1-\omega_2)t}\big].$$

For one color ($\omega_1 = \omega_2$) the cross term is steady and depends on the relative phase: the two beams interfere, and the two-photon rate is $F_{2p} = \kappa' |U_1 + U_2|^4$ [13], which is why two-beam 2PE shows fringes. For two colors the cross term oscillates at the difference frequency (57 THz, a 17.5 fs period for 750 and 656 nm) and averages away (Fig. 4). This holds only if the two spectra do not overlap [13]. In the vector case the cross term is $\mathrm{E}_1 \cdot \mathrm{E}_2^*$: beams polarized at right angles do not interfere at all, so polarization becomes a design variable the scalar papers cannot see.

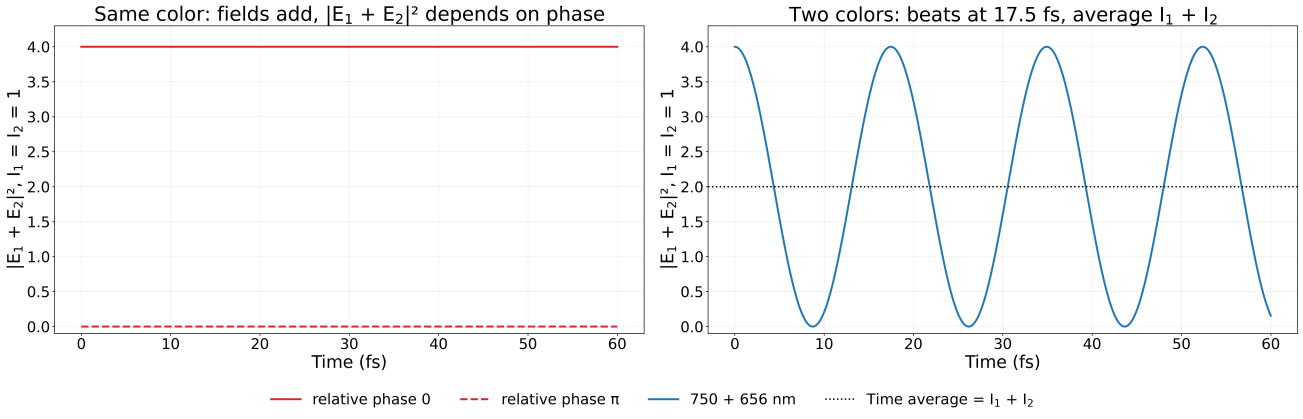


Figure 4: $|\mathrm{E}_1 + \mathrm{E}_2|^2$ over time for equal intensities. Same color: steady, and set by the relative phase (0 to 4). Two colors: beats at the difference frequency, averaging to $I_1 + I_2$.

9 Two beams at an angle

Placing the second lens at an angle θ to the first is the idea of theta microscopy [12]; $\theta = \pi$, two opposed lenses, is the 4Pi arrangement [11]. Beam 2 is computed in its own frame and rotated into the lab frame (Fig. 5); in the vector case its field vector, including its own E_z, rotates with it. Two cones of half-angle α meeting at $\theta = \pi/2$ fit only if $\alpha \le 45°$: NA $\le n/\sqrt{2}$, which is 0.71 at $n = 1$ ([13] p. 34) and 0.92 at $n = 1.3$.

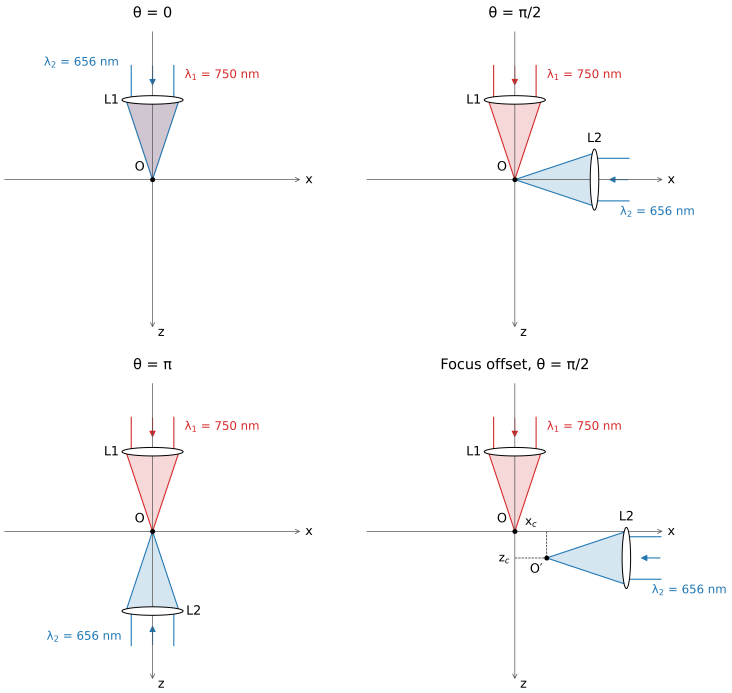


Figure 5: Geometry, after [13] Fig. 1 and [14] Fig. 1: beam 2 rotated by θ, focus moved to $O' = (x_c, z_c)$.

10 Optical pulses

A Gaussian pulse of intensity full width τ has a Gaussian power spectrum of full width $\Delta\nu = 2\ln 2/(\pi\tau) \approx 0.441/\tau$ [4, 16]. Its time-averaged focal intensity is the spectrum-weighted sum of monochromatic foci, $I(P) = \int S(\omega)\,|\mathrm{E}(P;\omega)|^2\,d\omega$ [16], which is what the current pulsed work computes. Two consequences for two-color excitation:

- At 1 fs the 750 nm and 656 nm spectra share 89% of their area (Fig. 6). The cross term of Section 8 no longer averages away, so few-cycle two-color excitation acquires interference. Lim & Saloma report spectral broadening negligible above 10 fs [15].
- Two-color excitation needs both pulses at the same place at the same time [15]. A 1 fs pulse is 0.3 µm long, shorter than the focal region, so a delay between the pulses moves where they meet: time and position couple.

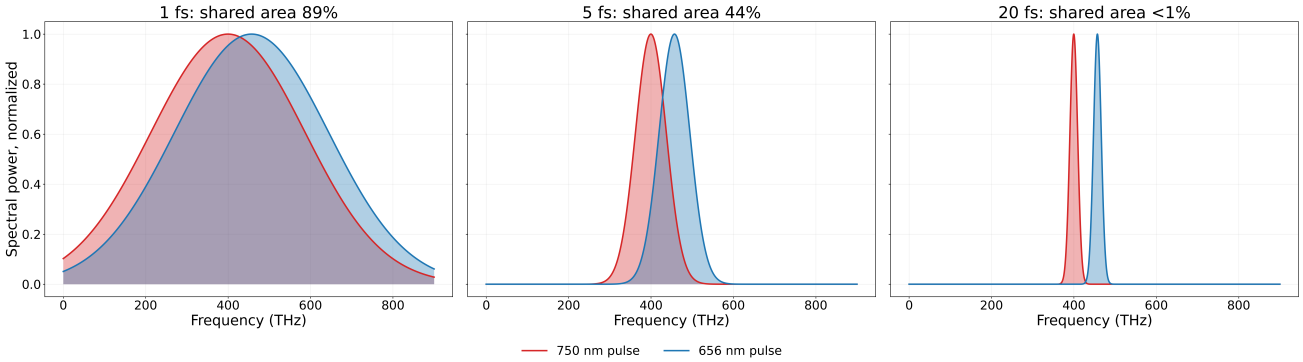


Figure 6: Power spectra of Gaussian pulses centered at 750 and 656 nm. At 1 fs they overlap almost completely; at 20 fs they are separate.

II. Replication at low NA

The three papers. Cambaliza & Saloma (2000) [13] introduces the comparison: 2CE against 2PE, circular against annular pupils, the wavelength gap $\Delta\lambda$, and a Monte Carlo scattering test, all with aligned beams. Lim & Saloma (2002) [14] moves beam 2's focus away from beam 1's. Lim & Saloma (2003) [15] keeps the foci together but gives each beam spherical aberration. Order done here: Lim 2002 first (it fixed the rotation convention), then Lim 2003 and Cambaliza 2000. All use the paraxial field of Section 6 at NA 0.5, $n = 1$, $\lambda_1 = 750$, $\lambda_2 = 656$ nm (2PE at 700 nm). Code: `mcdo/twobeam.py`, branch `two-color`; the field matches the Airy and sinc^2 profiles to 10^{-10}.

Result	Paper	Ours
Lim & Saloma 2002 — focus offset		
Peak loss, $\theta = \pi/2$, in plane	$-0.06x_c^2 + 1$; 11% at $1.33\lambda_1$	$-0.063x_c^2 + 0.999$; 11.1%
Peak loss, $\theta = \pi/2$, axial	$-0.045z_c^2 + 0.986$; 30% at $2.7\lambda_1$	$-0.045z_c^2 + 0.996$; 31.8%
Peak loss, $\theta = 0/\pi$, axial	$-0.027z_c^2 + 0.996$; 20% at $2.7\lambda_1$	$-0.027z_c^2 + 0.999$; 19.3%
Peak loss, $\theta = 0/\pi$, in plane	20% at $0.37\lambda_1$	17.7%
Linfoot F, $\theta = \pi/2$	stable in plane; ≈ -0.8 at $z_c = 1.33\lambda_1$	0.998; −0.90
Linfoot C, $\theta = \pi/2$	$\to 0$ at $z_c = 1.33\lambda_1$	1.01 *(not reproduced)*
Lim & Saloma 2003 — spherical aberration		
Peak compensation	along $\phi_1 = \phi_2$ ($\theta = 0$), $\phi_1 = -\phi_2$ (π)	0.84 of aligned at $\lvert\phi\rvert = 60$ on those lines
2CE fidelity, $\theta = \pi/2$	$F > 0.9$ for $\lvert\phi_{1,2}\rvert \le 40$	$F \ge 0.928$
Contour maps, Figs. 2–6	—	shapes reproduced (Fig. 7)
Cambaliza & Saloma 2000		
FWHM vs $\Delta\lambda$, $\theta = \pi/2$	data $\approx 0.715 \to 0.757\,\mu$m	$5.3 \times 10^{-4}\Delta\lambda + 0.713$
DOF vs $\Delta\lambda$, $\theta = \pi/2$	$-4.81 \times 10^{-4}\Delta\lambda + 0.711$	$-4.55 \times 10^{-4}\Delta\lambda + 0.713$
Annular 2CE vs circular 2PE, $\theta = 0$	30% narrower	30% narrower
Annular vs circular 2CE, $\theta = \pi/2$	32% narrower	29% narrower
Spot-size ratios (pp. 28–29)	0.67, 0.75, 1.58	0.49, 0.68, 2.98 *(not reproduced)*
Monte Carlo scattering, Fig. 8	—	not run (Monte Carlo shelved)

Notes. (i) Beam 2 must be rotated in real space; the optical-coordinate rotation written in [14] (p. 122) gives $-0.21x_c^2$ instead of $-0.06x_c^2$. (ii) The caption fit in [13] Fig. 5, FWHM $= 2.4 \times 10^{-4}\Delta\lambda + \ldots$, disagrees with its

own plotted points ($\approx 5\times10^{-4}$); ours follows the points. (iii) Cambaliza's spot-size ratios fail under FWHM, $1/e^2$ and first-minimum definitions, while its absolute widths in Fig. 5 match ours; the ratios are treated as unreliable. (iv) Lim 2002's C map: our Q, not C, falls to 0.05 at $z_c = 1.33\lambda_1$; the labels may be swapped. (v) With aberration at $\theta = 0$ the peak moves to $+z$ under the sign of [15] Eq. 1; the paper says "toward L1". To check.

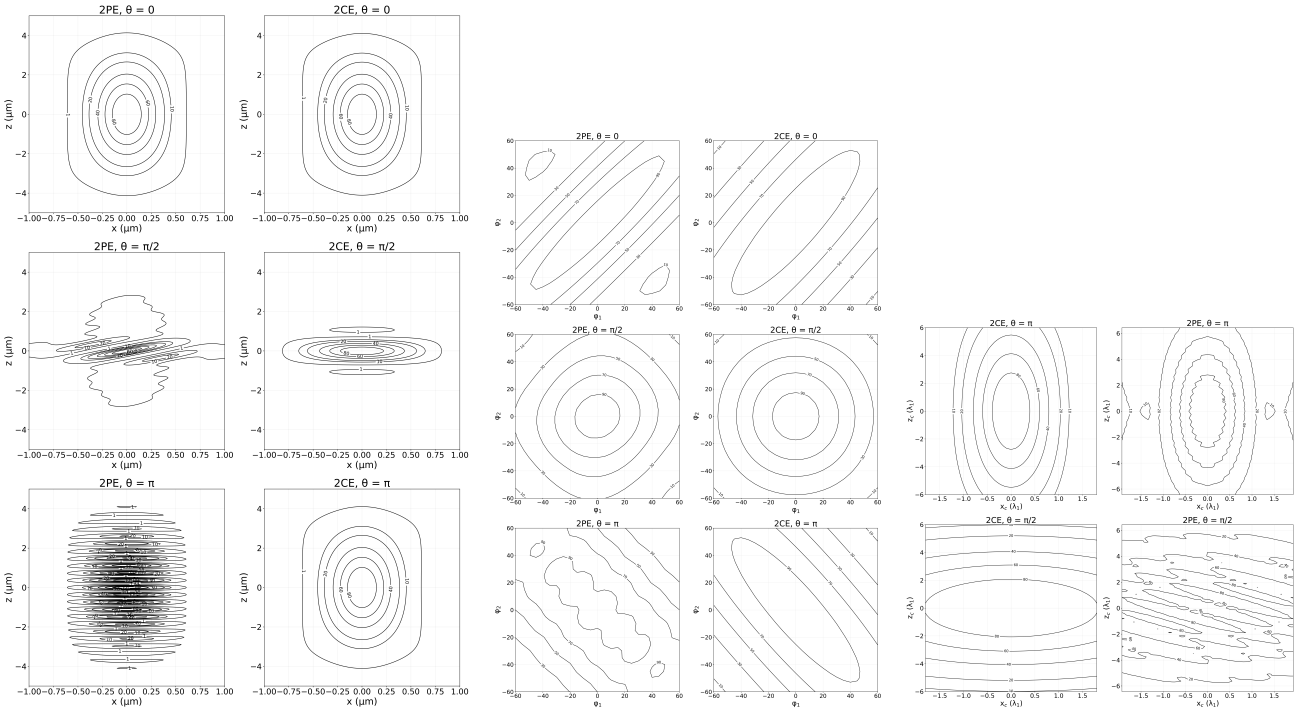


Figure 7: Left: aligned 2PE and 2CE point spread functions at $\theta = 0, \pi/2, \pi$, after [15] Fig. 2. Center: peak fluorescence (%) against ϕ_1, ϕ_2, after [15] Fig. 4. Right: peak fluorescence against focus offset, after [14] Fig. 4.

III. How it connects to the current work

- High NA. The Richards–Wolf engine (Section 5) replaces the paraxial field (Section 6). New choices appear that the scalar model hides: each beam's polarization, and E_z, which rotates with beam 2.
- Geometry. $\theta = \pi/2$ is limited to NA ≤ 0.92 at $n = 1.3$; beyond it only $\theta = 0, \pi$.
- Annular pupils. Cambaliza proposes long-DOF annular lenses to keep the foci overlapped ([13] p. 34). The extension law $(1 - \cos\alpha)/(\cos\vartheta_{\text{in}} - \cos\alpha)$ and the ε sweeps quantify that tolerance at high NA.
- Gaussian input. The truncation parameter α_t enters as each beam's $A(\vartheta)$.
- Pulses. The spectral integration of Section 10 is already built; at few-cycle durations the two-color spectra overlap and the no-interference assumption fails.
- Time delay. Needs the time-resolved field $\mathrm{E}(P, t)$; the engine currently gives the time-averaged intensity.
- Metrics. Linfoot F, C, Q are already in `mcdo.linfoot`.

Next steps

1. Resolve the open replication items: Lim 2002 C map; aberration sign.

2. The same geometry with the Richards–Wolf field at NA 0.5, choosing polarizations; report the scalar–vector difference.
3. Raise NA at $\theta = 0$ and π (and $\pi/2$ up to 0.92); add Gaussian input and annular pupils.
4. Pulsed two-color excitation: spectral overlap and interference at few-cycle durations.
5. Time delay between the pulses: time-resolved fields.

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